

Continuous R.V.'s:

Many quantities can take on an uncountably infinite # of values.

E.g.:

- Inches of snowfall
 - Duration of a call
 - Weight of a panda
- } Can be any non-neg. real number

If a random variable has an uncountably inf. range, it is called **continuous**.

To model these as r.v.'s, what is the prob. they equal a specific exact value?

E.g.: How likely am I to get 3.1415... inches of snow?

→ Zero! Why? Because an exact number (here, π) is too precise

An Example:

Say we want a r.v. X to have range of $[0, 1]$, and each value in its range is equally likely.

Goal: Show that $IP(X=x) = c > 0$, for all $x \in [0, 1]$ cannot occur.

Consider the following events:

$$\bullet A_1 = \{X = 1\}, A_2 = \{X = \frac{1}{2}\}, \dots A_k = \{X = \frac{1}{k}\}, \dots$$

Notice:

• $A_1, A_2, \dots$ are mutually exclusive $\Rightarrow$ Can use Axiom 3

• Their union is a strict subset of $[0, 1]$. (Recall: $U_{i=1}^{\infty} A_i \subset [0, 1]$, but $U_{i=1}^{\infty} A_i \neq [0, 1]$)

$$\Rightarrow IP(U_{i=1}^{\infty} A_i) \leq IP(S) = 1$$

By Axiom 3:

$$\begin{aligned} IP(U_{i=1}^{\infty} A_i) &= \sum_{i=1}^{\infty} IP(A_i) \\ &= \sum_{i=1}^{\infty} c \quad \text{= } c \text{ by assumption} \end{aligned}$$

$$IP(U_{i=1}^{\infty} A_i) = \infty ? \quad \text{↯ } c > 0$$

→ Should be at most 1... $\Rightarrow$ We must have $c = 0!$

Since $IP(X=x)=0$ for all x if X is a continuous R.V.,
 we can't characterize these r.v.'s by a pmf.
 We need another tool - looking not at points, but intervals

Def. (Probability Density Function): For a cts. r.v. X , its
 probability density function, pdf, is whatever function $f_X()$

that satisfies:

$$IP(a \leq X \leq b) = \int_a^b f_X(x) dx,$$

for any two $a \leq b$.

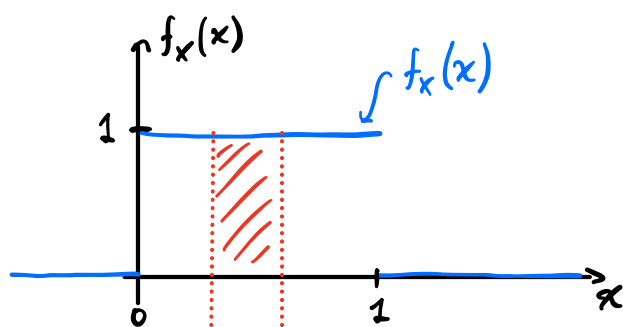
Properties of pdf:

- $f_X(x) \geq 0$, for all x [Non-negative] (Q: Can $f_X(x)$ be larger than 1?)
- $\int_{-\infty}^{\infty} f_X(x) dx = 1$ [Integrates to 1]

Note: $IP(X=x) \neq f_X(x)$, the former is always zero if X is a cts. r.v.

Illustration of pdf:

A r.v. with range $[0,1]$, and equally likely values:

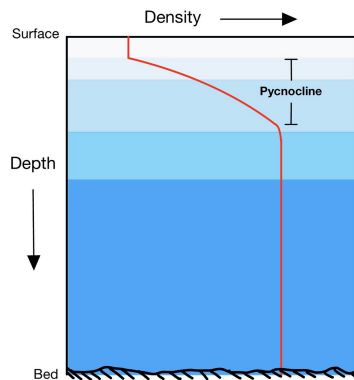


$$0.3 \quad 0.6 \quad \leadsto IP(0.3 \leq X \leq 0.6) = 1 \cdot (0.6 - 0.3) = \underline{0.3}$$

Observe that the total area under $f_X(x)$ is equal to 1!

Intuition:

The density of water in our oceans depends on depth:

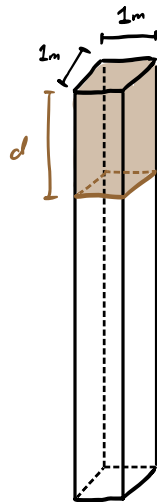


Density: Units of kg/m^3

Depth: Units of m

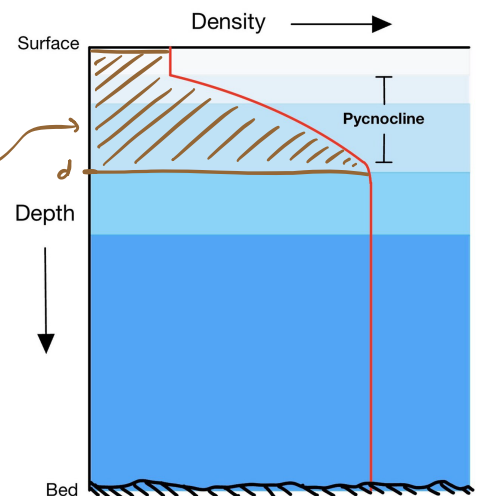
Mass: Units of kg

To get the mass $[\text{kg}]$ of water in a $1 \times 1 \text{m}$ column at various depths, we'd need to integrate density $[\text{kg}/\text{m}^3]$ over volume $[\text{m}^3]$.



$$\text{Mass} = 1\text{m} \times 1\text{m} \cdot \int_0^d \text{Density}(\text{Depth}) \cdot d(\text{Depth})$$

Equals this shaded area



Note: If we have zero volume, then the mass is always zero, regardless of the density.

↪ Analogous to $P(X=x)=0$, since the interval $[x,x]$ has length of zero

Idea: Think of probability as mass

If the units of a cts. r.v. X are in meters, then

the units of $f_X(\cdot)$ are in probability/meter

⇒ Units of $\int_a^b f_X(x) dx$ are in probability!

$[\text{Prob.}/\text{m}] \quad \uparrow \quad \uparrow \quad [\text{m}]$

Observation: What happens if we look at $P(x-\varepsilon \leq X \leq x+\varepsilon)$ for very small values of $\varepsilon > 0$?

By def'n: $P(x-\varepsilon \leq X \leq x+\varepsilon) = \int_{x-\varepsilon}^{x+\varepsilon} f_X(u) du$ (def'n of pdf)

$$\approx f_X(x) \cdot (x+\varepsilon - (x-\varepsilon))$$

$$\underline{P(x-\varepsilon \leq X \leq x+\varepsilon) \approx 2\varepsilon \cdot f_X(x) = [\text{Units of } X] \cdot [\text{Probability / Units of } X]}$$

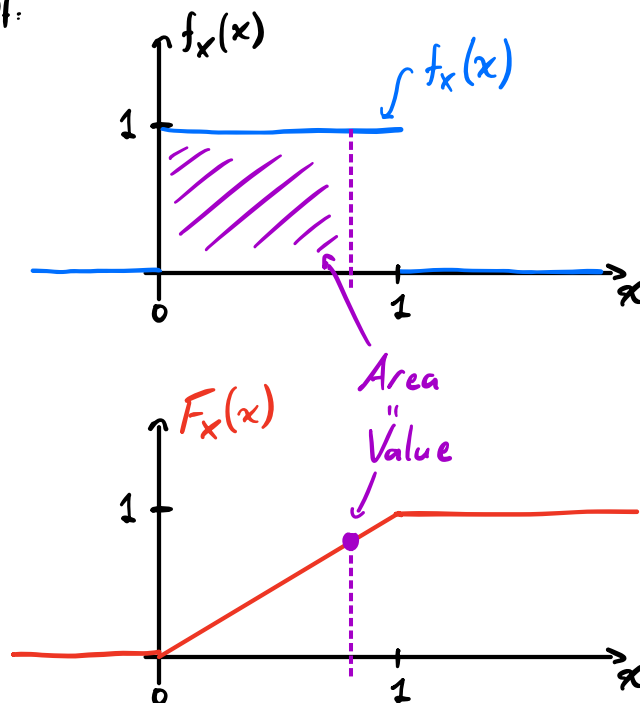
$\leadsto$ So, $f_X(x)$ still tells us about how likely X is to be near x .

Although discrete / cts r.v.'s differ in our ability to define a mass / density function, they both share the same def'n of a cdf.

Def. (cdf, continuous): For a cts r.v. X with density $f_X(\cdot)$, its cumulative distribution / density function, $F_X(\cdot)$, equals

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(u) du, \text{ for all } x$$

Illustration of cdf:



$\leadsto$ Unlike the discrete case, this cdf has no jumps! (i.e. is continuous)

Properties of (DF):

Same as
for discrete
cdf's

- $0 \leq F_X(x) \leq 1$
- If $a \leq b$, then $F_X(a) \leq F_X(b)$ [Non-decreasing]
- Proof: Same as the discrete case
- As $x \rightarrow \infty$, $F_X(x) \rightarrow 1$

• If $F_X(x)$ is differentiable at x , then its derivative equals the pdf.
$$\frac{dF_X(x)}{dx} = f_X(x)$$

Note: The cdf is not always differentiable. See above ex. at $x=0$.

Expected Value & Variance:

Like before, we want to compute expectations of (functions of) a random variable. However, since $P(X=x)=0$ for all x if X is a cts. r.v., we have to define the expectation differently.

Def. (Expectation, cts.): For a r.v. X with pdf $f_X(x)$, and any real-valued function $g(\cdot)$, we have:

$$E[X] = \int_{-\infty}^{\infty} x \cdot f_X(x) dx, \quad E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Note: It is still true that $E[a \cdot X + b] = a \cdot E[X] + b$ for constants a, b , even if X is a continuous r.v.

Def. (Variance, cts.): The variance of X is $Var(X) = \sigma_X^2 = E[(X - E[X])^2]$, its standard deviation is $\sigma_X = \sqrt{Var(X)}$.

Properties: Same as before

- $Var(X) = E[X^2] - E[X]^2$
- $Var(a \cdot X + b) = a^2 \cdot Var(X)$

~> Same as discrete case! Only difference is in how we compute $E[\cdot]$.

Discrete $\Rightarrow$ Described by pmf $\Rightarrow E[\cdot]$ is a sum

Cts. $\Rightarrow$ Described by pdf $\Rightarrow E[\cdot]$ is an integral

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A Digression...

It is actually possible to have r.v.'s that are
both discrete and continuous. E.g. Let $X \sim \text{Ber}(1/2)$, and
 Y be any continuous r.v.

What is the dist'n of the following random variable?

$$Z = \begin{cases} 0 & \text{if } X=0 \\ Y & \text{if } X=1 \end{cases}$$

$\rightarrow$ Equals 0 w/ prob. $1/2 \dots$

$\rightarrow$ Equals any value in $\text{Range}(Y)$ w/ prob. 0...

~> Its range is uncountably infinite but it has $P(Z=0)=1/2$?

$\Rightarrow$ Continuous R.V.?

$\Rightarrow$ Discrete R.V.?

In this class we won't see such things...

Curious about how we might deal with them?

$\Rightarrow$ Take more advanced probability courses!

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Common Continuous Distributions:

Same idea as before - we want to group cts. random variables into families, identify the parameters that vary within a family, and study the properties of each r.v. in the family.

Uniform Distribution:

Def: We say that $X \sim \text{Unif}(a, b)$, for $a < b$, if its pdf equals:

$$f_X(x; a, b) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{o/w} \end{cases}$$

Parameters: $a, b \in \mathbb{R}$, and $a < b$. Specify the interval of interest.

Range: $[a, b]$

Note: If $a=0$, $b=1/2$, then $f_X(x; 0, 1/2) = \begin{cases} 2 & \text{if } 0 \leq x \leq 1/2 \\ 0 & \text{o/w} \end{cases}$

Hence, the pdf can be greater than one.

Interpretation/Use: Not much to say here... But, notice that if X is a cts. r.v., then the random variable $F_X(X)$ has a $\text{Unif}[0, 1]$ distribution!

I.e.: Define the r.v. $Y := F_X(X)$, and let's try to find the cdf of Y .

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(F_X(X) \leq y) \\ &= P(X \leq F_X^{-1}(y)) \quad \begin{array}{l} \text{Take } F_X^{-1}(\cdot) \text{ on both sides} \\ \text{def. of } F_X(\cdot) \end{array} \\ &\stackrel{=}{=} F_X(F_X^{-1}(y)) = y \quad \text{for any } y \in [0, 1] \end{aligned}$$

Properties:

$$\bullet E[X] = \frac{1}{2}(b+a)$$

$$\bullet \text{Var}(X) = \frac{1}{12}(b-a)^2$$

Exponential Distribution:

Def: We say $X \sim \text{Exp}(\lambda)$, if its pdf is:

$$f_X(x; \lambda) = \begin{cases} \lambda \cdot e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{o/w} \end{cases}$$

Its cdf is given by: $F_X(x; \lambda) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{o/w} \end{cases}$

Parameter: $\lambda > 0$, called the "rate" Range: $[0, \infty)$

Interpretation/Use: A waiting time, similar to a geometric r.v., but taking on any positive real value.

This distribution has a neat connection to Poisson r.v.'s:

Let $X_1, X_2, \dots$ are indep. $\text{Exp}(\lambda)$ random variables.

The 1st "event" occurs at time X_1 , the 2nd "event" at time $X_1 + X_2$, the 3rd at $X_1 + X_2 + X_3$, and so on.

Given any $T > 0$, interpreted as some window of time,

then $\sum_{k=1}^{\infty} \mathbb{1}\{\sum_{i=1}^k X_i \leq T\}$ will have a $\text{Poi}(\lambda \cdot T)$ distribution!
= # of events that occurred before time T .

Proof? $\leadsto$ ORF309...

Properties:

- $E[X] = 1/\lambda$
 - $\text{Var}(X) = 1/\lambda^2$
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Normal/Gaussian Distribution:

Def: We say $X \sim N(\mu, \sigma^2)$ if its pdf equals

$$f_X(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \text{ for } x \in \mathbb{R}.$$

If $\mu=0$ and $\sigma^2=1$, we say that X has a standard normal distribution, whose cdf is denoted/defined as:

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du, \text{ for } x \in \mathbb{R}. \quad \parallel \text{ We can't compute this analytically, so we use tables/software...}$$

Parameters: $\mu \in \mathbb{R}$, the mean/expected value of X
 $\sigma^2 > 0$, the variance of X .

Range: $\mathbb{R}$, any real number

Interpretation/Use: Appears all the time - we will see why in a few weeks... More on this soon!

Properties:

- $E[X] = \mu$, $\text{Var}(X) = \sigma^2$
- If $X \sim N(\mu, \sigma^2)$, then for $a \in \mathbb{R}, b > 0$, $\frac{X-a}{b}$ has a $N(\frac{\mu-a}{b}, \frac{\sigma^2}{b^2})$ distribution!

Note: It is not special that $E[\frac{X-a}{b}] = \frac{\mu-a}{b}$, and $\text{Var}(\frac{X-a}{b}) = \frac{\sigma^2}{b^2}$.

What is unique here: that $\frac{X-a}{b}$ has a normal distribution!

$\Rightarrow$ So, $\frac{X-\mu}{\sigma} \sim N(0,1)$ is always a standard normal r.v.

$$\begin{aligned} \Rightarrow P(a \leq X \leq b) &= P\left(\frac{a-\mu}{\sigma} \leq \frac{X-\mu}{\sigma} \leq \frac{b-\mu}{\sigma}\right) \\ &= P\left(\frac{X-\mu}{\sigma} \leq \frac{b-\mu}{\sigma}\right) - P\left(\frac{X-\mu}{\sigma} \leq \frac{a-\mu}{\sigma}\right) \end{aligned}$$

Std. normal Std. normal

$$P(a \leq X \leq b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

Use Axiom 3 with $A_1 = \{X \leq a\}$, $A_2 = \{a \leq X \leq b\}$

- If $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$ are independent, then $X+Y \sim N(\mu_x+\mu_y, \sigma_x^2+\sigma_y^2)$. More on this later - what is special here is again that the resulting dist'n is normal...

- Preview (+ why this distribution is so common):

If $X_1, \dots, X_n$ are all independent r.v.'s with identical distribution,

(e.g. $X_1, \dots, X_n$ are all $\text{Ber}(1/2)$), then the quantity $\frac{1}{n} \sum_{i=1}^n X_i$

has approximately a normal distribution, where:

- as n grows, the approximation is better
- the exact distribution of X_i (and $X_2, \dots, X_n$) does not matter

→ We will say more about this later. The power of this result is that the distribution of the r.v.'s $X_1, \dots, X_n$ becomes irrelevant!

These are not the only families of continuous distributions we will encounter in the course...

However, they are very common, and will suffice for now...

In general, should you encounter another family of distributions, it is always good to ask:

- What is its pmf/pdf, and cdf?
 - What are the parameters, what do they represent?
 - What is the range of such a r.v.?
 - What is the interpretation/use of a r.v. with this distribution?
 - What are its properties (e.g. mean/variance)?
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Summary of Discrete vs. Ctc. R.V.'s:

	$ Range(X) $	Characterized By	Computing Expectation
Continuous	Uncountably Infinite	Prob. of falling in an interval $\leadsto$ <u>density</u> fn.	An <u>integral</u> over <u>density</u> multiplied by value.
Discrete	Finite or Countably Infinite	Prob. of equaling a value $\leadsto$ <u>mass</u> fn.	A <u>sum</u> over <u>mass</u> times value

A Useful Feature of the Uniform Dist'n:

Let X be an $\text{Exp}(2)$ random variable, whose cdf is given by:

$$F_X(x) = 1 - e^{-2x}, \text{ for } x \geq 0.$$

Recall that the CDF of a $\text{Unit}(0,1)$ r.v. (call it Y) is equal to:

$$F_Y(y) = y, \text{ for } y \in [0,1]$$

Goal: Show that the random variable $1 - e^{-2 \cdot X}$ has a $\text{Unit}(0,1)$ distribution.

We do this by computing $\mathbb{P}(1 - e^{-2X} \leq u)$ for all u , and showing that this equals the cdf of a $\text{Unit}(0,1)$ r.v.

First, notice that since $X \sim \text{Exp}(2)$, then $\text{Range}(X) = [0, \infty)$.

$\leadsto$ Hence, the random variable " $1 - e^{-2X}$ " has range of $[0,1]$

$$\begin{aligned} \mathbb{P}(1 - e^{-2X} \leq u) &= \mathbb{P}(-e^{-2X} \leq u - 1) \quad (\text{Subtract 1 from both sides}) \\ (\text{for } u \in [0,1]) &= \mathbb{P}(e^{-2X} \geq 1 - u) \quad (\text{Multiply by } -1 \text{ on both sides}) \\ &= \mathbb{P}(-2X \geq \log(1 - u)) \quad (\text{Take } \log(\cdot) \text{ of both sides}) \\ &= \mathbb{P}(X \leq -\frac{1}{2} \log(1 - u)) \quad (\text{Divide by } -2) \end{aligned}$$

$\uparrow$ We know the distribution of X ! It is $\text{Exp}(2)$.

So, $\mathbb{P}(X \leq c)$ can just be expressed as $F_X(c)$!

$$\begin{aligned} &= 1 - \exp(-2 \cdot \underbrace{(-\frac{1}{2} \log(1 - u))}_{\text{cancel}}) \quad (\text{By knowing CDF of } X) \\ &= 1 - \exp(\underbrace{\log(1 - u)}_{\text{cancel}}) \end{aligned}$$

$$\mathbb{P}(1 - e^{-2X} \leq u) = 1 - (1 - u) = u, \text{ for } u \in [0,1] \quad \parallel \text{CDF of a } \text{Unit}(0,1) \text{ r.v.}!$$

Conclusion: $1 - e^{-2X}$ has a $\text{Unit}(0,1)$ distribution when $X \sim \text{Exp}(2)$!

Why does this matter?

→ It means that we can build a r.v. with any distribution if:

i) We know the cdf of the r.v. we want

ii) we can generate r.v.'s from the $\text{Unif}(0,1)$ distribution.

Using the same process as above, you can check that

if $Y \sim \text{Unif}(0,1)$, then " $-\frac{1}{2} \log(1-Y)$ " has an $\text{Exp}(2)$ dist'n!

Pf: Since $Y \in [0,1]$, then $-\frac{1}{2} \log(1-Y) \in [0, \infty)$.

For any $t \in [0, \infty)$:

$$\mathbb{P}\left(-\frac{1}{2} \log(1-Y) \leq t\right) = \mathbb{P}(\log(1-Y) \geq -2t)$$

$$= \mathbb{P}(1-Y \geq e^{-2t})$$

$$= \mathbb{P}(Y \leq 1 - e^{-2t})$$

↓ Since Y is $\text{Unif}(0,1)$, $\mathbb{P}(Y \leq c) = c$ for any $c \in [0,1]$

$$\underline{\underline{\mathbb{P}\left(-\frac{1}{2} \log(1-Y) \leq t\right) = 1 - e^{-2t}}}$$

⇒ So the r.v. " $-\frac{1}{2} \log(1-Y)$ " has an $\text{Exp}(2)$ dist'n!

This (remarkably) holds in general.

• If X has cdf $F_X(\cdot)$, then $F_X(X)$ is $\text{Unif}(0,1)$ distributed

• If $Y \sim \text{Unif}(0,1)$, then $F_X^{-1}(Y)$ is a random variable w/
cdf $F_X(\cdot)$.

This is very useful computationally!
